

Extremal no-three-in-line subsets of a modular hyperbola in the Hall–Jackson–Sudbery–Wild window

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Abstract

Let p be an odd prime, $c \in \mathbb{F}_p^*$, and let $H(c, p) = \{(x, y) \in \mathbb{Z}^2 : xy \equiv c \pmod{p}\}$ be the modular hyperbola. Hall, Jackson, Sudbery and Wild (1975) observed that the $2p \times 2p$ window $G(p) = [-\frac{p-1}{2}, \frac{3p-1}{2}] \times [0, 2p-1]$ contains exactly four points of each of the $p-1$ residue classes of $H(c, p)$, at the vertices of a $p \times p$ square, and that keeping three of the four (the twelve outer blocks T_2 of the 4×4 block subdivision of $G(p)$) yields $3(p-1)$ points with no three collinear. Kovács, Nagy and Szabó (2025/26) state that this deletion is optimal, i.e. that $H(c, p) \cap T_2$ is a largest no-three-in-line subset of $H(c, p) \cap G(p)$. We give a short self-contained proof of this statement for every c , and we determine the complete extremal structure: we classify all lines carrying three or more points of $H(c, p) \cap G(p)$ (they have slope ± 1 , and their number is $\frac{3}{2}(p-1) - s$), we exhibit an explicit linear-programming certificate for the bound $3(p-1)$ (so even the fractional relaxation has value $3(p-1)$), and we show that the number of maximum no-three-in-line subsets is exactly 9^s , where $s \in \{0, 1, 2\}$ is the number of quadratic residues among c and $-c$. In particular the HJSW set is the unique maximum if and only if $p \equiv 1 \pmod{4}$ and c is a quadratic non-residue. Everything is verified by an independent program for all primes $p \leq 101$.

1 Introduction

The no-three-in-line problem asks for the largest subset of the $n \times n$ grid with no three collinear points; at most $2n$ points are possible, and the best general lower bound, $\frac{3}{2}n - o(n)$, is due to Hall, Jackson, Sudbery and Wild [1]: for $n = 2p$, p prime, they place $3(p-1)$ points of a modular hyperbola $H(c, p)$ in the window $G(p)$. Their construction rests on two facts: every line meets $H(c, p)$ in at most two residue classes, and $G(p)$ contains exactly four points of each class, at the vertices of a $p \times p$ square; keeping the three copies outside the middle block of the 4×4 block subdivision of $G(p)$ leaves no three collinear.

Kovács, Nagy and Szabó [2] recently used the same window and its refinements for the no- $(k+1)$ -in-line problem. In their setting the HJSW set is the case $|W| = 1$, $k = 2$ of a general randomised construction (all four copies of every class are taken, then a minimum set R of points is deleted so that every line of slope ± 1 retains at most k points), and they state that for $|W| = 1$, $k = 2$ the HJSW deletion “essentially details an optimal choice” of the deleted points (v1, 2025), resp. yields the “largest no-3-in-line set $S_2(c, p)$ inside $S(W)$ ” (v2, 2026). The general estimate of [2, Rem. 3.4], $|R| \geq \frac{1}{2}X$ with $X = \sum_{\ell} (|\ell \cap S(W)| - k)^+$, gives for $|W| = 1$, $k = 2$ the bound $|R| \geq p - 1 - s$ (see Remark 8 below), i.e. $|S| \leq 3(p-1) + s$ for every no-three-in-line $S \subseteq H(c, p) \cap G(p)$: exact when neither c nor $-c$ is a square modulo p , but one or two short of the truth otherwise. We are not aware of a proof of the exact statement in [2]; the purpose of this note is to give one, and to determine the whole extremal structure of the problem.

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Results. Throughout, p is an odd prime, $h = (p - 1)/2$, $c \in \mathbb{F}_p^*$, and

$$H(c, p) = \{(x, y) \in \mathbb{Z}^2 : xy \equiv c \pmod{p}\}, \quad G(p) = \{-h, \dots, 3h + 1\} \times \{0, \dots, 2p - 1\},$$

and $\mathcal{P} := H(c, p) \cap G(p)$. A set is *lawful* (no-three-in-line, in general position) if it contains no three collinear points. A line is *rich* if it contains at least three points of $\mathcal{P}$; $\mathcal{L}$ denotes the set of rich lines. Let $s = [c \text{ is a QR}] + [-c \text{ is a QR}] \in \{0, 1, 2\}$ (QR = quadratic residue modulo p).

Theorem 1. (i) Every rich line has slope $+1$ or -1 ; there are exactly $\frac{3}{2}(p - 1) - s$ rich lines, $p - 1$ of them carrying three points of $\mathcal{P}$ and $\frac{1}{2}(p - 1) - s$ carrying four (Proposition 7 lists them); exactly $2s$ points of $\mathcal{P}$ lie on no rich line.

(ii) Every lawful subset of $\mathcal{P}$ has at most $3(p - 1)$ points; the bound is attained. Even the fractional relaxation (maximise $\sum_q x_q$ subject to $\sum_{q \in \ell} x_q \leq 2$ for $\ell \in \mathcal{L}$, $0 \leq x_q \leq 1$) has optimum $3(p - 1)$.

(iii) The number of lawful subsets of size $3(p - 1)$ is exactly 9^s . Every maximum set coincides with the HJSW set $H(c, p) \cap T_2$ outside the (at most four) classes $(a, \pm a)$ with $a^2 = \pm c$.

Corollary 2. If $p \equiv 3 \pmod{4}$ then $s = 1$ for every c : there are exactly 9 maximum sets. If $p \equiv 1 \pmod{4}$ then $s = 2$ for c a QR (81 maximum sets) and $s = 0$ for c a non-residue: the HJSW set is the unique maximum. Thus $H(c, p) \cap T_2$ is the unique largest lawful subset of $H(c, p) \cap G(p)$ if and only if $p \equiv 1 \pmod{4}$ and c is a quadratic non-residue.

The proof is a double count over the rich lines: for a lawful S , $|S| \leq Z + 2|\mathcal{L}|$ where Z is the number of points on no rich line, and the identity $Z + 2|\mathcal{L}| = 3(p - 1)$ follows from the classification of the rich lines. The count is exact precisely because the $2s$ points that lie on no rich line are accounted separately; this is the correction to the averaging bound of [2] mentioned above. All rich lines lie inside orbits of a Klein four-group acting on the classes, and the equality analysis reduces to two small “gadgets” (Figure 1), which gives 9^s .

2 Notation

Two points are *congruent* if their coordinates agree modulo p ; a *class* is a congruence class, identified with an element of $\mathbb{F}_p \times \mathbb{F}_p$. Points of $H(c, p)$ have both coordinates prime to p , so the classes of $H(c, p)$ are the $p - 1$ pairs $\kappa = (a, b) \in \mathbb{F}_p^* \times \mathbb{F}_p^*$ with $ab = c$. Since each coordinate range of $G(p)$ is an interval of $2p$ consecutive integers, it contains exactly two representatives of every residue class; hence every class κ of $H(c, p)$ has exactly four points in $G(p)$ and $|\mathcal{P}| = 4(p - 1)$. We write (x_κ, y_κ) for the *base copy* of κ , its unique point in $\{-h, \dots, h\} \times \{1, \dots, p - 1\}$ (necessarily $x_\kappa \neq 0$), and

$$\kappa(r, s) := (x_\kappa + rp, y_\kappa + sp), \quad r, s \in \{0, 1\},$$

for its four points in $G(p)$. Following [1, 2] we sort the classes into four *types* according to the position of the base copy:

$$A : x_\kappa > 0, y_\kappa > h; \quad B : x_\kappa < 0, y_\kappa > h; \quad C : x_\kappa > 0, y_\kappa \leq h; \quad D : x_\kappa < 0, y_\kappa \leq h.$$

(In the notation of [2, Def. 2.3]: $\kappa(0, 0) \in A_{0,0}, B_{-1,0}, C_{0,0}, D_{-1,0}$ respectively; the HJSW set T_2 omits, for each type, the copy $A_{0,0}, B_{0,0}, C_{0,1}, D_{0,1}$, i.e. $\kappa(0, 0)$ for A , $\kappa(1, 0)$ for B , $\kappa(0, 1)$ for C , $\kappa(1, 1)$ for D ; the union of these omitted blocks is the middle block M of [2, §2.3]. We call this copy the *middle copy* of κ .) We write n_A, n_B, n_C, n_D for the numbers of classes of each type, so $n_A + n_B + n_C + n_D = p - 1$. Finally, for a class κ put

$$d_\kappa := x_\kappa - y_\kappa, \quad e_\kappa := x_\kappa + y_\kappa, \quad D_\kappa := \{x - y = d_\kappa\}, \quad E_\kappa := \{x + y = e_\kappa + p\};$$

D_κ (the *diagonal* of κ) contains $\kappa(0,0), \kappa(1,1)$ and E_κ (the *antidiagonal*) contains $\kappa(0,1), \kappa(1,0)$. For a real number x let $x^+ = \max\{x, 0\}$.

Three involutions act on the classes of $H(c, p)$:

$$\sigma(a, b) = (-b, -a), \quad \tau(a, b) = (b, a), \quad \nu(a, b) = (-a, -b) = \sigma\tau(a, b) = \tau\sigma(a, b),$$

so $V = \{1, \sigma, \tau, \nu\}$ is a Klein four-group acting on the $p-1$ classes. σ has fixed classes iff $-c$ is a QR (then exactly two: $(a, -a)$ with $a^2 = -c$), τ has fixed classes iff c is a QR (two: (a, a) with $a^2 = c$), and ν has none. Put $f_\sigma := \#\{\kappa : \sigma\kappa = \kappa\} \in \{0, 2\}$, $f_\tau := \#\{\kappa : \tau\kappa = \kappa\} \in \{0, 2\}$; thus $f_\sigma + f_\tau = 2s$. The V -orbits are of size 4 (*generic*: four distinct classes) or of size 2 (*exceptional*: $\{\kappa, \nu\kappa\}$ with κ and $\nu\kappa$ both fixed by σ , or both fixed by τ); there are exactly s exceptional orbits.

3 Rich lines

Lemma 3. *Let ℓ be a line containing at least two points of $H(c, p)$. Then the points of $H(c, p) \cap \ell$ lie in at most two classes. More precisely, if ℓ contains a point of class κ , then the classes met by ℓ are: only κ if ℓ has slope 0 or ∞ ; among $\{\kappa, \sigma\kappa\}$ if ℓ has slope $+1$; among $\{\kappa, \tau\kappa\}$ if ℓ has slope -1 .*

Proof. Let $(u, v) \in \mathbb{Z}^2$ be a primitive direction vector of ℓ and $(x_0, y_0) \in H(c, p) \cap \ell$. The lattice points of ℓ are $(x_0 + tu, y_0 + tv)$, $t \in \mathbb{Z}$, and such a point lies on $H(c, p)$ iff $uv t^2 + (x_0 v + y_0 u) t \equiv 0 \pmod{p}$ (we used $x_0 y_0 \equiv c$). The class of the point depends only on $t \pmod{p}$. If $uv \not\equiv 0$, the congruence has at most two roots modulo p ; if exactly one of u, v is divisible by p (both cannot be, as $\gcd(u, v) = 1$), it reduces to $x_0 v t \equiv 0$ or $y_0 u t \equiv 0$ with a unit coefficient, so $t \equiv 0$: one class. For $(u, v) = (1, 0), (0, 1)$ this gives the claim for slopes 0, ∞ . For $(u, v) = (1, 1)$ the roots are $t \equiv 0$ and $t \equiv -(x_0 + y_0)$, the latter giving the point $(-y_0, -x_0)$ modulo p , of class $\sigma\kappa$. For $(u, v) = (1, -1)$ the roots are $t \equiv 0$ and $t \equiv y_0 - x_0$, giving (y_0, x_0) , of class $\tau\kappa$. $\square$

Lemma 4. *Two distinct points of one class in $G(p)$ differ by $\pm(p, 0), \pm(0, p), \pm(p, p)$ or $\pm(p, -p)$; the line through them has slope 0, $\infty, +1$ or -1 and contains no third point of that class. The point $\kappa(r, s)$ lies on the slope- $(+1)$ line $x - y = d_\kappa + (r - s)p$ and on the slope- (-1) line $x + y = e_\kappa + (r + s)p$. In particular D_κ and E_κ are the only lines of slope ± 1 carrying two points of κ .*

Proof. The four points of a class are the vertices of an axis-parallel square of side p ; no three are collinear. The rest is a direct computation. $\square$

Corollary 5. *A rich line has slope $+1$ or -1 . Consequently a set $S \subseteq \mathcal{P}$ is lawful iff $|S \cap \ell| \leq 2$ for every $\ell \in \mathcal{L}$.*

Proof. By Lemma 3 three points of $\mathcal{P}$ on a line lie in at most two classes, so two of them are congruent, and by Lemma 4 the line has slope 0, $\infty, \pm 1$. A line of slope 0 or ∞ meets a single class (Lemma 3), which has at most two points on it (Lemma 4). $\square$

Lemma 6 (position of the partner classes). *For every class κ the types of $\sigma\kappa, \tau\kappa$ and the shifts $d_{\sigma\kappa} - d_\kappa, e_{\tau\kappa} - e_\kappa$ are given by the following table:*

type of κ	type of $\sigma\kappa$	$d_{\sigma\kappa} - d_\kappa$	type of $\tau\kappa$	$e_{\tau\kappa} - e_\kappa$
A	A	0	D	$-p$
B	C	$+p$	B	0
C	B	$-p$	C	0
D	D	0	A	$+p$

In particular σ preserves the types A, D and interchanges B, C ; τ preserves B, C and interchanges A, D ; ν interchanges $A \leftrightarrow D$ and $B \leftrightarrow C$; the fixed classes of σ lie in $A \cup D$, those of τ in $B \cup C$; and $n_A = n_D, n_B = n_C$.

Proof. Write $\kappa = (a, b)$ with base copy (x_κ, y_κ) , so $x_\kappa \equiv a, y_\kappa \equiv b$. The base copy of $\sigma\kappa = (-b, -a)$ has first coordinate $-y_\kappa$ if $y_\kappa \leq h$ and $p - y_\kappa$ if $y_\kappa > h$, and second coordinate $p - x_\kappa$ if $x_\kappa > 0$ and $-x_\kappa$ if $x_\kappa < 0$ (these are the representatives of $-b$ in $[-h, h]$ and of $-a$ in $[1, p-1]$). Going through the four types: $\kappa \in A$: $(p - y_\kappa, p - x_\kappa)$, again of type A , difference $x_\kappa - y_\kappa$; $\kappa \in D$: $(-y_\kappa, -x_\kappa)$, type D , difference $x_\kappa - y_\kappa$; $\kappa \in C$: $(-y_\kappa, p - x_\kappa)$, type B , difference $x_\kappa - y_\kappa - p$; $\kappa \in B$: $(p - y_\kappa, -x_\kappa)$, type C , difference $x_\kappa - y_\kappa + p$. Similarly, the base copy of $\tau\kappa = (b, a)$ has first coordinate y_κ if $y_\kappa \leq h$ and $y_\kappa - p$ otherwise, and second coordinate x_κ if $x_\kappa > 0$ and $x_\kappa + p$ otherwise: $\kappa \in B$: $(y_\kappa - p, x_\kappa + p)$, type B , sum e_κ ; $\kappa \in C$: (y_κ, x_κ) , type C , sum e_κ ; $\kappa \in A$: $(y_\kappa - p, x_\kappa)$, type D , sum $e_\kappa - p$; $\kappa \in D$: $(y_\kappa, x_\kappa + p)$, type A , sum $e_\kappa + p$. The remaining assertions follow. $\square$

Proposition 7. $\mathcal{L}$ consists of the following lines, and no others; for each we list its points in $\mathcal{P}$.

- (a) For $\kappa \in A \cup D$ with $\sigma\kappa \neq \kappa$: the common diagonal $D_\kappa = D_{\sigma\kappa}$, with the four points $\kappa(0, 0), \kappa(1, 1), \sigma\kappa(0, 0), \sigma\kappa(1, 1)$.
- (b) For $\kappa \in B \cup C$ with $\tau\kappa \neq \kappa$: the common antidiagonal $E_\kappa = E_{\tau\kappa}$, with the four points $\kappa(0, 1), \kappa(1, 0), \tau\kappa(0, 1), \tau\kappa(1, 0)$.
- (c) For every $\kappa \in C$: D_κ with the three points $\kappa(0, 0), \kappa(1, 1), \sigma\kappa(1, 0)$ ($\sigma\kappa \in B$); for every $\kappa \in B$: D_κ with the three points $\kappa(0, 0), \kappa(1, 1), \sigma\kappa(0, 1)$ ($\sigma\kappa \in C$).
- (d) For every $\kappa \in A$: E_κ with the three points $\kappa(0, 1), \kappa(1, 0), \tau\kappa(1, 1)$ ($\tau\kappa \in D$); for every $\kappa \in D$: E_κ with the three points $\kappa(0, 1), \kappa(1, 0), \tau\kappa(0, 0)$ ($\tau\kappa \in A$).

The lines listed in (a)–(d) are pairwise distinct (apart from the identifications $D_\kappa = D_{\sigma\kappa}$ in (a) and $E_\kappa = E_{\tau\kappa}$ in (b) that are built into the list), and each of them lies inside a single V -orbit of classes. Consequently

$$|\mathcal{L}| = \frac{n_A + n_D - f_\sigma}{2} + \frac{n_B + n_C - f_\tau}{2} + (n_A + n_D) + (n_B + n_C) = \frac{3}{2}(p-1) - s, \quad (1)$$

with $p-1$ three-point lines and $\frac{1}{2}(p-1) - s$ four-point lines. Moreover, the points of $\mathcal{P}$ lying on no rich line are exactly $\kappa(1, 1)$ for σ -fixed $\kappa \in A$, $\kappa(0, 0)$ for σ -fixed $\kappa \in D$, $\kappa(0, 1)$ for τ -fixed $\kappa \in B$ and $\kappa(1, 0)$ for τ -fixed $\kappa \in C$; their number is

$$Z := \#\{q \in \mathcal{P} : q \text{ lies on no rich line}\} = f_\sigma + f_\tau = 2s. \quad (2)$$

The points lying on two rich lines are exactly the middle copies of the classes κ whose partner ($\sigma\kappa$ for $\kappa \in A \cup D$, $\tau\kappa$ for $\kappa \in B \cup C$) is different from κ ; no point lies on three rich lines.

Proof. Let $\ell \in \mathcal{L}$. By Corollary 5 its slope is ± 1 ; say $+1$ (the case -1 is symmetric, with τ, e_κ, E_κ in place of $\sigma, d_\kappa, D_\kappa$). By Lemma 3 the points of $\ell \cap \mathcal{P}$ lie in two classes $\kappa, \sigma\kappa$ (a single class carries at most two points on a line), and one of the classes, say κ , has two points on ℓ ; by Lemma 4 these are $\kappa(0, 0), \kappa(1, 1)$ and $\ell = D_\kappa$, and $\sigma\kappa \neq \kappa$. The points of $\sigma\kappa$ on D_κ are the $\sigma\kappa(r, s)$ with $d_{\sigma\kappa} + (r-s)p = d_\kappa$. By Lemma 6: for $\kappa \in A \cup D$, $r = s$, giving (a); for $\kappa \in C$, $r - s = 1$, giving $\sigma\kappa(1, 0)$; for $\kappa \in B$, $r - s = -1$, giving $\sigma\kappa(0, 1)$; this is (c). Conversely each listed line has at least three points.

Distinctness: if $D_\kappa = D_\lambda$ then this line of slope $+1$ meets both classes κ, λ , so $\lambda \in \{\kappa, \sigma\kappa\}$ by Lemma 3; and $D_\kappa = D_{\sigma\kappa}$ holds for $\kappa \in A \cup D$ (shift 0) but not for $\kappa \in B \cup C$ (shift $\pm p$). Likewise $E_\kappa = E_\lambda$ forces $\lambda \in \{\kappa, \tau\kappa\}$, and $E_\kappa = E_{\tau\kappa}$ holds exactly for $\kappa \in B \cup C$. A rich line meets only

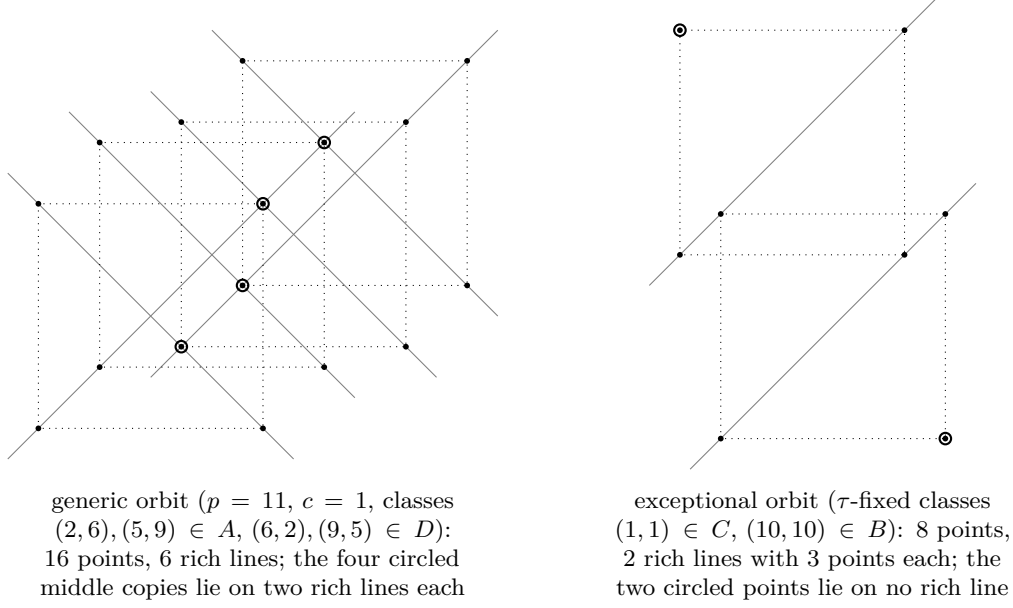


Figure 1: The two gadgets of the equality analysis, drawn to scale for $p = 11, c = 1$ (dotted squares: the four copies of a class; grey: rich lines). Generic orbit: the circled points are excluded and the other 12 are forced ($= T_2$). Exceptional orbit: the circled points are forced and each rich line keeps two of its three points, $3 \cdot 3 = 9$ choices.

the classes $\kappa, \sigma\kappa$ (or $\kappa, \tau\kappa$), which lie in one V -orbit. Counting: (a) gives $(n_A + n_D - f_\sigma)/2$ lines, (b) $(n_B + n_C - f_\tau)/2$, (c) $n_B + n_C$, (d) $n_A + n_D$, which is (1).

For the incidences of a single point, take $\kappa \in A$. Its points $\kappa(0, 1), \kappa(1, 0)$ lie on $E_\kappa \in \mathcal{L}$ (d), and on no line of type (a)/(c) (their slope-(+1) lines $x - y = d_\kappa \pm p$ carry no second point of κ and, by the table, no point of $\sigma\kappa$). $\kappa(1, 1)$ lies on D_κ , which is rich iff $\sigma\kappa \neq \kappa$, and on $x + y = e_\kappa + 2p = e_{\tau\kappa} + 3p$, which carries no point of $\tau\kappa$: so $\kappa(1, 1)$ is on one rich line if $\sigma\kappa \neq \kappa$ and on none otherwise. $\kappa(0, 0)$ lies on D_κ and on $x + y = e_\kappa = e_{\tau\kappa} + p$, i.e. on $E_{\tau\kappa} \in \mathcal{L}$: two rich lines if $\sigma\kappa \neq \kappa$, one otherwise. The types D, B, C are analogous (for B : $\kappa(1, 0) \in D_{\sigma\kappa}$ since $d_\kappa + p = d_{\sigma\kappa}$, while $\kappa(0, 1)$ lies on $x - y = d_\kappa - p = d_{\sigma\kappa} - 2p$, which carries no point of $\sigma\kappa$; for C the roles of $(1, 0), (0, 1)$ are exchanged; for D : $\kappa(1, 1) \in E_{\tau\kappa}$ since $e_\kappa + 2p = e_{\tau\kappa} + p$, and $\kappa(0, 0)$ lies on $x + y = e_\kappa = e_{\tau\kappa} - p$). This gives (2) and the last two assertions. $\square$

4 Proof of Theorem 1 and Corollary 2

Part (i) is Corollary 5 and Proposition 7.

Upper bound. Let $S \subseteq \mathcal{P}$ be lawful. For $q \in \mathcal{P}$ let $m(q)$ be the number of rich lines through q . Every rich line carries at most two points of S (Corollary 5), hence

$$\begin{aligned}
 |S| &= \#\{q \in S : m(q) = 0\} + \#\{q \in S : m(q) \geq 1\} \leq Z + \sum_{q \in S} m(q) - \sum_{q \in S} (m(q) - 1)^+ \\
 &= Z + \sum_{\ell \in \mathcal{L}} |S \cap \ell| - \sum_{q \in S} (m(q) - 1)^+ \leq Z + 2|\mathcal{L}|,
 \end{aligned} \tag{3}$$

and by (1), (2), $Z + 2|\mathcal{L}| = 2s + 3(p - 1) - 2s = 3(p - 1)$. The HJSW set attains the bound [1].

LP certificate. The same count is valid for fractional $x \in [0, 1]^{\mathcal{P}}$ with $\sum_{q \in \ell} x_q \leq 2$ ($\ell \in \mathcal{L}$): $\sum_q x_q \leq \sum_{q: m(q)=0} x_q + \sum_{\ell \in \mathcal{L}} \sum_{q \in \ell} x_q \leq Z + 2|\mathcal{L}|$. In LP-duality terms the dual solution assigns weight 1 to every rich line and weight 1 to the bound $x_q \leq 1$ of each of the Z points on no rich line. This proves (ii).

Equality. $|S| = 3(p - 1)$ holds iff all inequalities in (3) are equalities, i.e. iff (1) S contains all Z points lying on no rich line; (2) $|S \cap \ell| = 2$ for every $\ell \in \mathcal{L}$; (3) no point of S lies on two rich lines. Since every rich line lies inside one V -orbit of classes (Proposition 7), conditions (1)–(3) split into independent conditions on the orbits, and the number of maximum sets is the product of the numbers of admissible choices in the orbits. In a generic orbit $\{\kappa, \sigma\kappa, \tau\kappa, \nu\kappa\}$ all four classes have distinct partners; the six rich lines meeting the orbit (two of type (a) or (b), four of type (c) or (d)) meet no other orbit, its four middle copies have $m = 2$ and are excluded by (3), and each of the six lines then retains exactly two points, both with $m = 1$, which are forced into S by (2): on a generic orbit S coincides with $H(c, p) \cap T_2$, and there is exactly one admissible choice. An exceptional orbit $\{\kappa, \nu\kappa\}$ (κ and $\nu\kappa$ fixed by σ , or both fixed by τ) meets exactly two rich lines, each with three points of $m = 1$, and contains two points with $m = 0$: conditions (1)–(3) allow $\binom{3}{2}\binom{3}{2} = 9$ choices, and all of them are lawful by Corollary 5. There are s exceptional orbits, whence 9^s maximum sets, all agreeing with the HJSW set outside the exceptional orbits, i.e. outside the classes $(a, \pm a)$ with $a^2 = \pm c$. This proves (iii).

Corollary 2. -1 is a QR modulo p iff $p \equiv 1 \pmod{4}$. If $p \equiv 3 \pmod{4}$, exactly one of $c, -c$ is a QR, so $s = 1$; if $p \equiv 1 \pmod{4}$, c and $-c$ have the same character, so $s \in \{0, 2\}$ according as c is a non-residue or a residue. $\square$

Remark 8. (a) In the language of [2, Constr. 3.2, Rem. 3.4] with $W = \{c\}$, $k = 2$ (so $S(W) = \mathcal{P}$ and L' is the set of lines of slope ± 1): $X_{2,p}(\{c\}) = \sum_{\ell} (|\ell \cap \mathcal{P}| - 2)^+ = (p-1) \cdot 1 + (\frac{1}{2}(p-1) - s) \cdot 2 = 2(p-1) - 2s$ by Theorem 1(i), so the general bound $|R| \geq \frac{1}{2}X$ gives $|R| \geq p-1-s$ and hence, for every lawful $S \subseteq \mathcal{P}$, $|S| \leq |\mathcal{P}| - |R| \leq 3(p-1) + s$ (a lawful S is a trimming, by Corollary 5). The averaging loses exactly one point per exceptional orbit, because each such orbit contains two points on no rich line and no point on two rich lines; the count (3) keeps track of these points and is exact.

(b) The theorem says that no lawful set can use the middle block M profitably: adding fourth copies to the HJSW set is impossible without removing at least as many points. Any improvement of the HJSW construction must therefore use points outside $H(c, p) \cap G(p)$ (another hyperbola, another window, or non-hyperbola points).

(c) The proof uses only: at most two classes per line (Lemma 3), the square arrangement of the four copies, and the position of the partner classes (Lemma 6). It does not use $p \geq 7$; $p = 3$ and $p = 5$ are covered as well.

5 Independent computer verification

The theorem has a short self-contained proof; the following check is an independent sanity test of the bookkeeping (the classification of the rich lines, the exceptional classes and the count of the maximum sets), performed by a program that trusts no part of the proof. The script `slack/hjsw_window_check.py` of the repository recomputes, from the point set alone: all lines with at least three points of $\mathcal{P}$ (found by brute force over all pairs of points; all have slope ± 1 , three or four points, and lie inside one V -orbit); the identity $2|\mathcal{L}| + Z = 3(p-1)$; the exact maximum by exhaustive search inside each orbit (at most 2^{16} subsets), assembling a maximum witness which is passed through the independent verifier `saturation.the_law` in grid coordinates; and the number of maximum sets, compared with 9^s . All checks pass for every prime $3 \leq p \leq 101$ (all $c \in \mathbb{F}_p^*$ for $p \leq 31$; $c \in \{1, 2, 3, -1, -2, (p+1)/2\}$ for larger p); log: `bench/hjsw_window_check_all.log`. The values agree with the exact maxima obtained earlier in the project by SAT solving (kissat/CaDiCaL, two encodings, witnesses through the law): 18, 30, 36, 48, 54, 66 for $p = 7, 11, 13, 17, 19, 23$. An external referee report on v0.1 of this note (kept in `docs/reviews/`) re-derived the point sets and the rich lines with an independent implementation and confirmed $3(p-1)$ and 9^s for all c with $p \leq 31$ and for samples of c with $p = 37, 43, 59, 73, 101$.

Repository (public): <https://github.com/iwasborninbali/saturation>, tag `paper-v0.5`; the verification script and its log were committed as `d422b78` (2026-08-17); the note itself is `paper/hjsw_window.tex`.

p	c	$ \mathcal{P} $	$ \mathcal{L} $	Z	max	#max sets
7	1	24	8	2	18	9
11	1	40	14	2	30	9
13	1	48	16	4	36	81
13	2	48	18	0	36	1
17	1	64	22	4	48	81
19	1	72	26	2	54	9
23	1	88	32	2	66	9
101	1	400	148	4	300	81

Table 1: Sample values from the verification log ($|\mathcal{L}| = \frac{3}{2}(p-1) - s$, $Z = 2s$).

AI/tool-use disclosure. The theorem, its proof, the verification program and the text of this note were produced by an autonomous large-language-model agent (Anthropic Claude, model *Claude Fable 5*, August 2026), working in a repository shared with two earlier agents of the same kind, which had established the computational observations (exact maxima by SAT for $p \leq 23$) and the setup; the human author posed the question, organised the work, obtained an external referee report, and takes responsibility for the statements made here. All code, logs and the correspondence of the agents are available in the repository.

References

- [1] R. R. Hall, T. H. Jackson, A. Sudbery, K. Wild, Some advances in the no-three-in-line problem, *J. Combin. Theory Ser. A* 18 (1975) 336–341.
- [2] B. Kovács, Z. L. Nagy, D. R. Szabó, Randomised algebraic constructions for the no- $(k+1)$ -in-line problem, *arXiv:2508.07632* (v1 11 Aug 2025, v2 21 Jul 2026); *Advances in Combinatorics* 2026:7.